



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 17: General Equilibrium Models

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ICS2523 Microeconomics

Lecture 17

Last lecture we

- Did a comparative statics analysis of the long-run equilibrium
- Determined the producer surplus in the long-run

In today's lecture, we will finish **Competitive Markets**. We will

- Learn about welfare analysis in the long-run
- Recall some behavioural assumptions

We will move to **General Equilibrium**

- General Equilibrium with Two Goods $\rightarrow$ Edgeworth Box

Review Encuesta de Evaluación Temprana

Economic Efficiency and Welfare Analysis

Producer Theory

7. Economic Efficiency and Welfare Analysis

Long-run competitive equilibrium may have the property of allocating resources *efficiently*

Concepts that we need to remember

- Consumer surplus
- Producer surplus

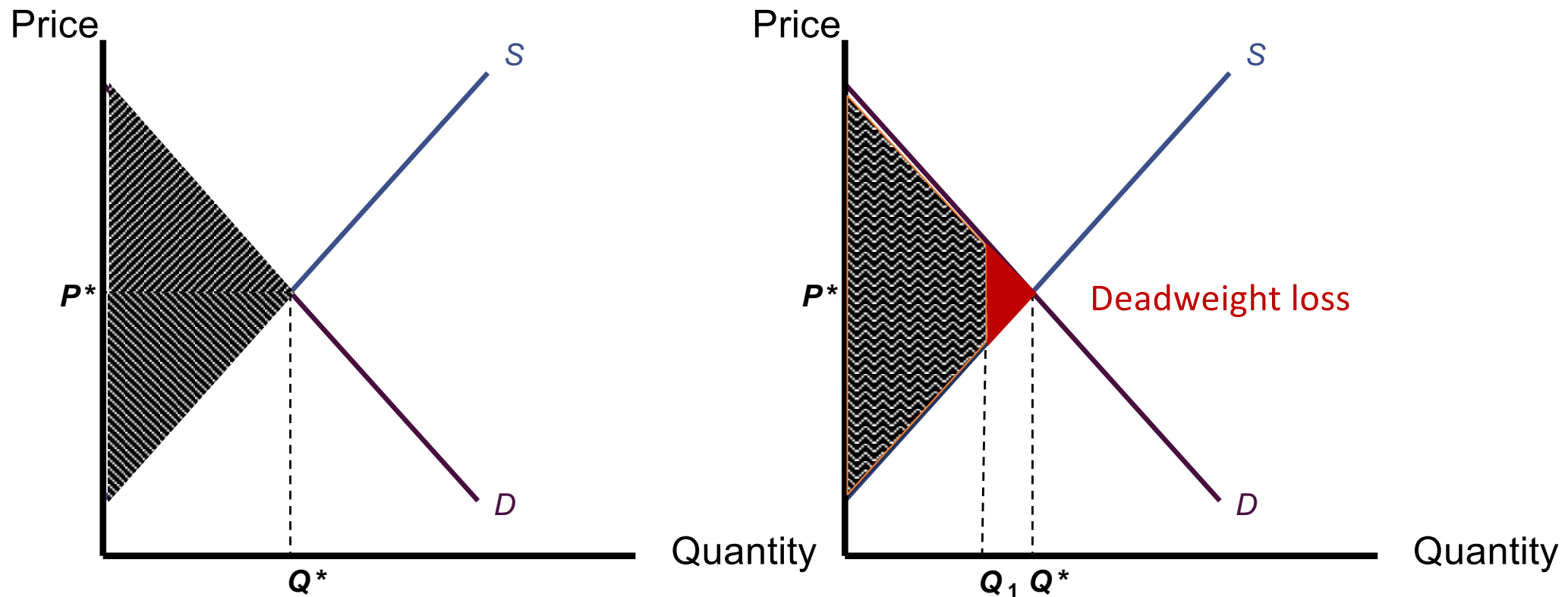
The area between the demand curve and supply curve represents the sum of consumer and producer surplus → represents the total additional value obtained by market participants

What happens at the competitive market equilibrium?

This area is MAXIMIZED!

7. Economic Efficiency and Welfare Analysis

Graphic proof



7. Economic Efficiency and Welfare Analysis

Mathematical proof

Consumer surplus + Producer surplus

$$= [U(Q) - PQ] + \left[PQ - \int_0^Q P(Q) dQ \right]$$

$$= U(Q) - \int_0^Q P(Q) dQ$$

In the LR equilibrium $\rightarrow P(Q) = AC = MC$

If we maximize the above equation with respect to Q yields:

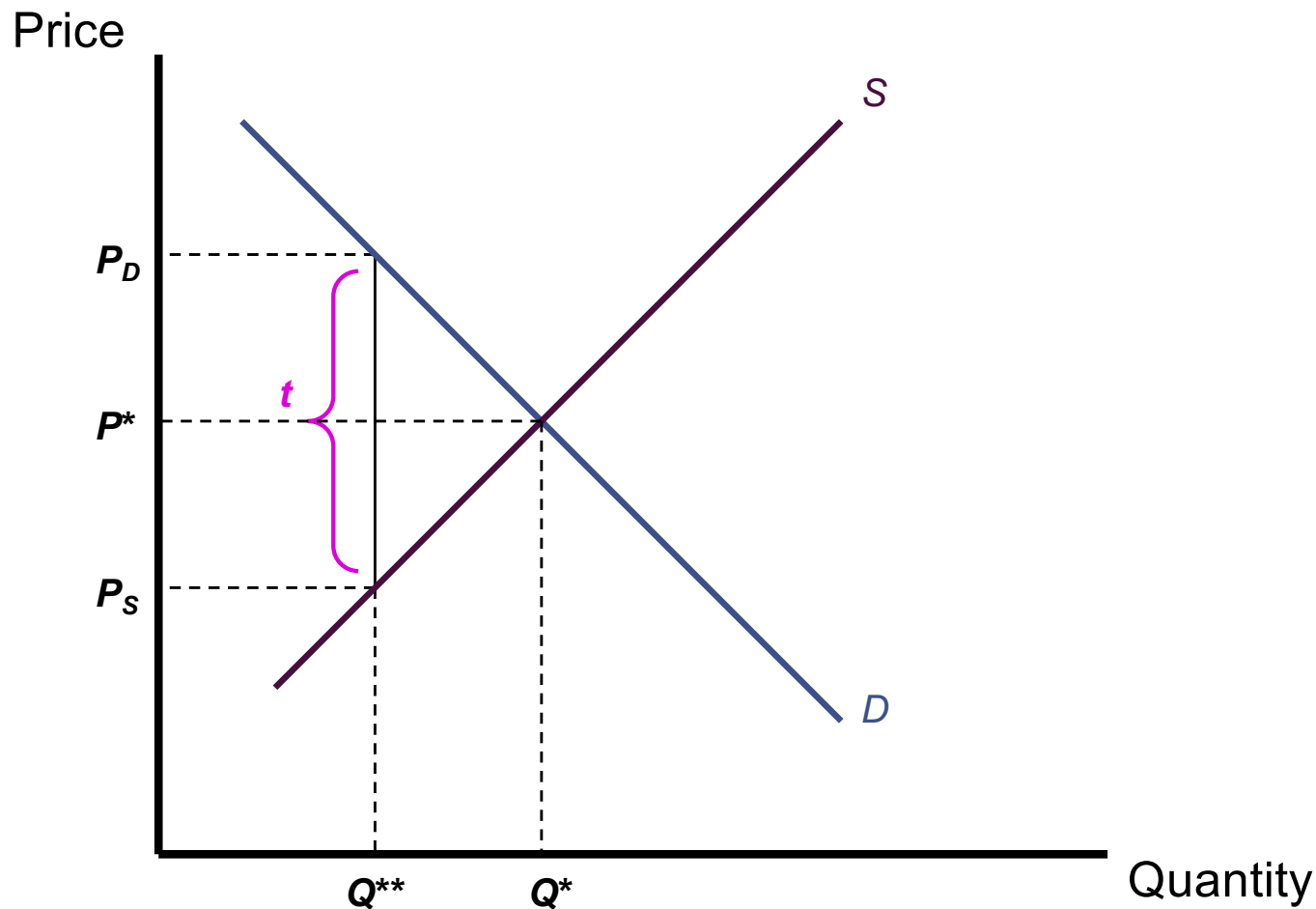
$$U'(Q) = P(Q)$$

$$\rightarrow U'(Q) = P(Q) = AC = MC$$

$\rightarrow$ Maximization occurs when the marginal value of Q to the representative consumer is equal to market price

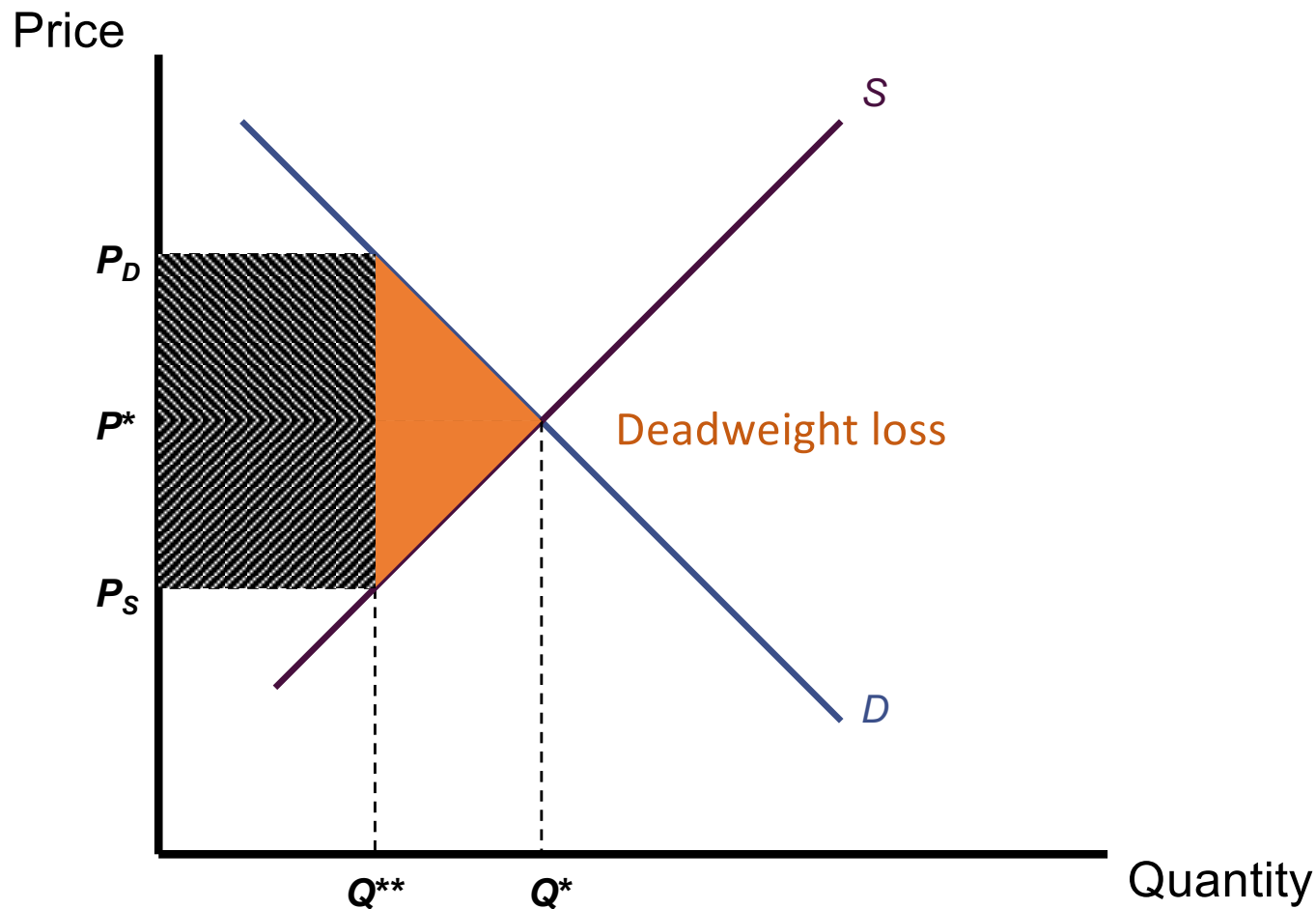
7. Economic Efficiency and Welfare Analysis

Example: Tax incidence analysis $\rightarrow P_D - P_S = t$ (per-unit tax)



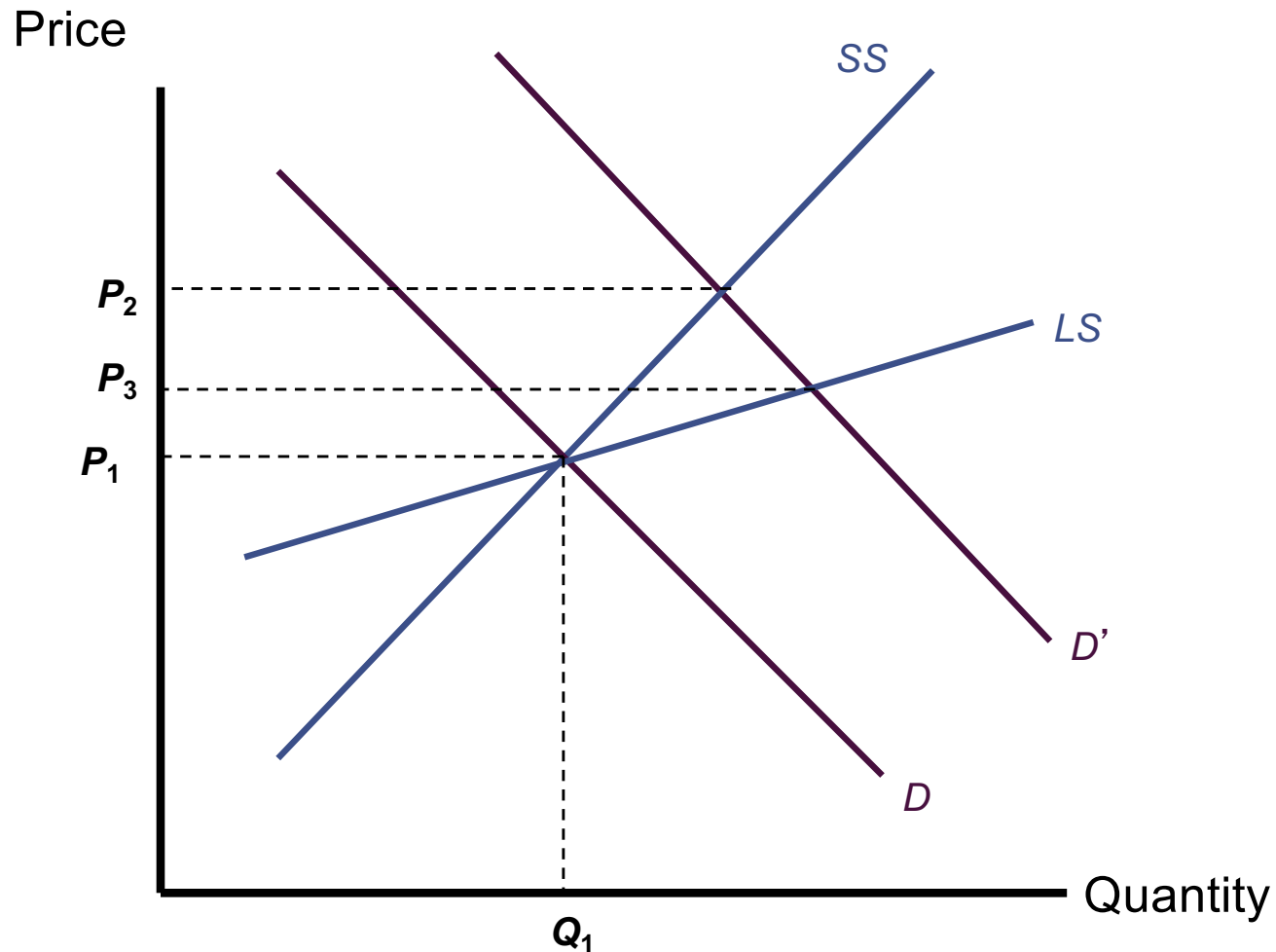
7. Economic Efficiency and Welfare Analysis

Example: Tax incidence analysis $\rightarrow P_D - P_S = t$ (per-unit tax)



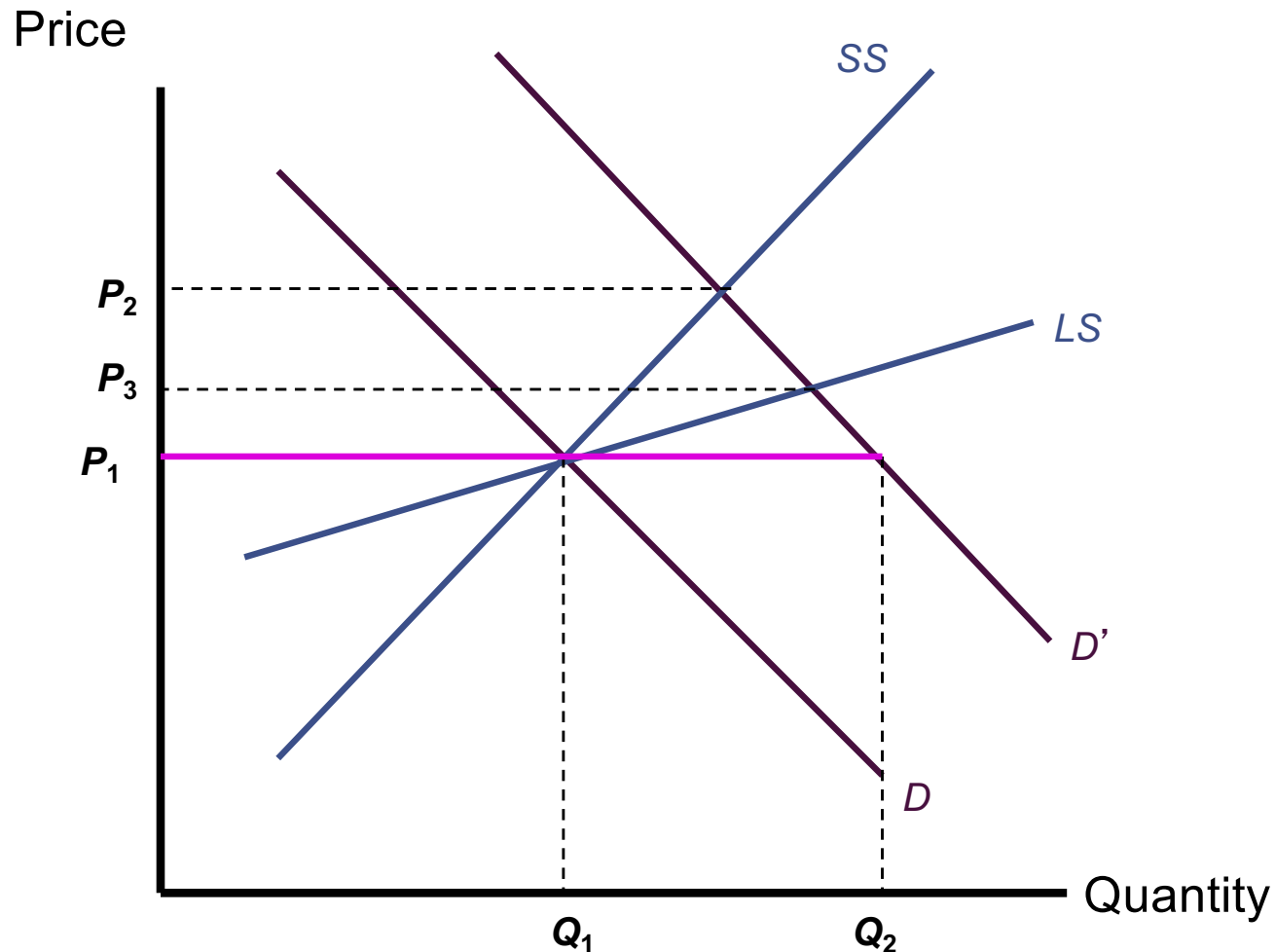
7. Economic Efficiency and Welfare Analysis

Example: Price controls and shortages



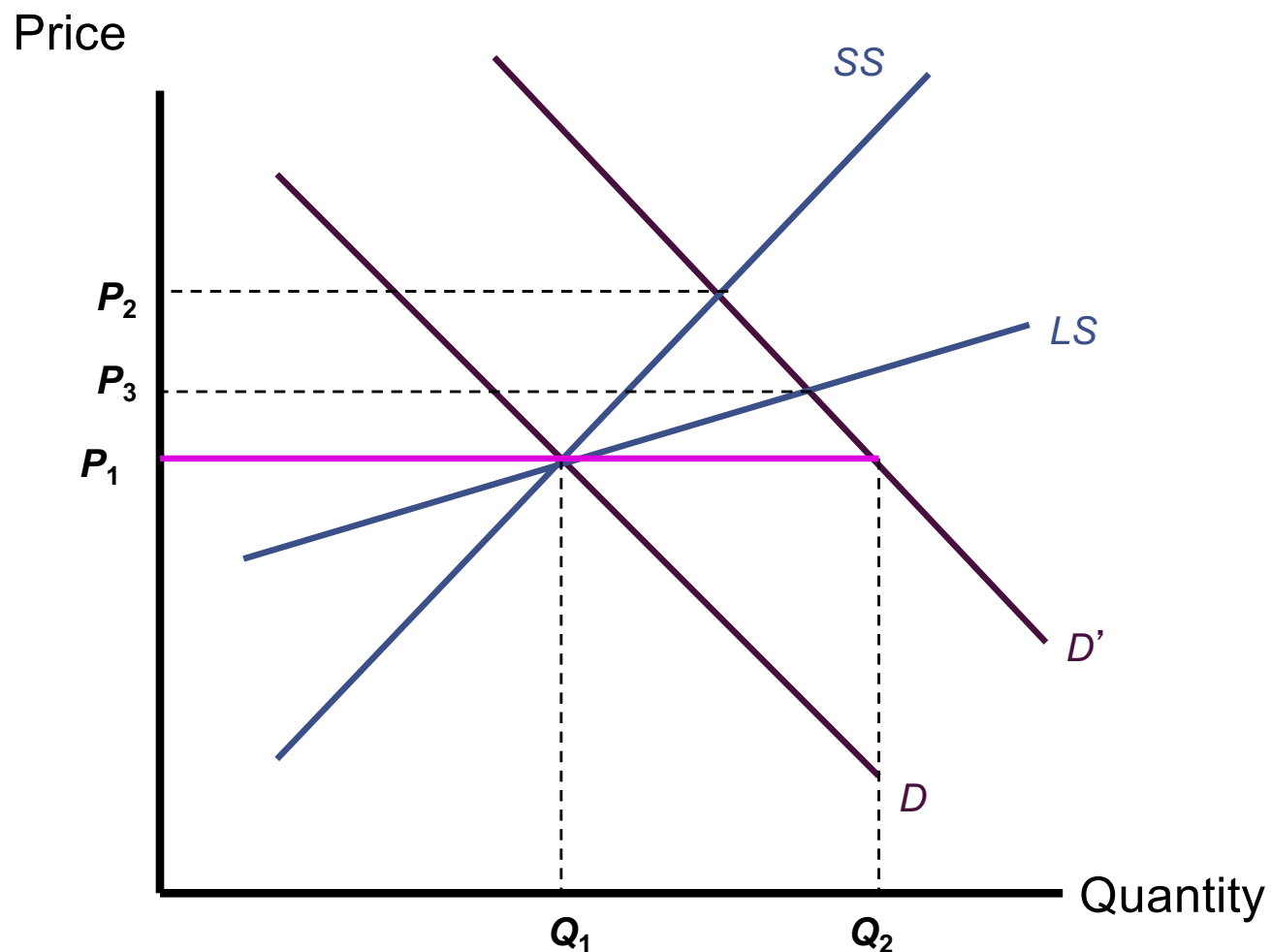
7. Economic Efficiency and Welfare Analysis

Example: Price controls and shortages



Activity 1

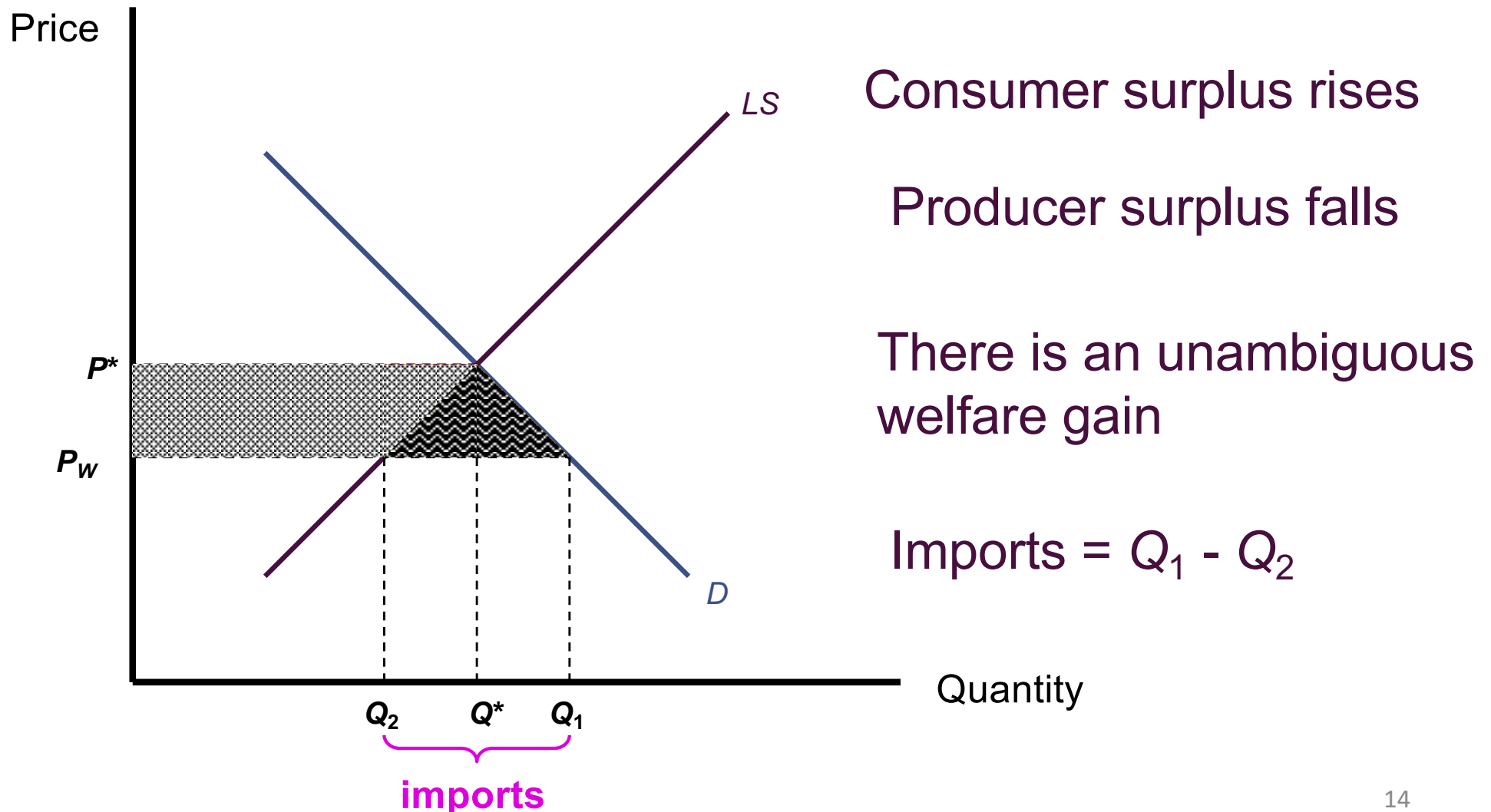
Analyze gains and losses by the use of a ceiling price (in this case).



Identify who gains, who losses, and how the welfare changes (we are better or worse?)

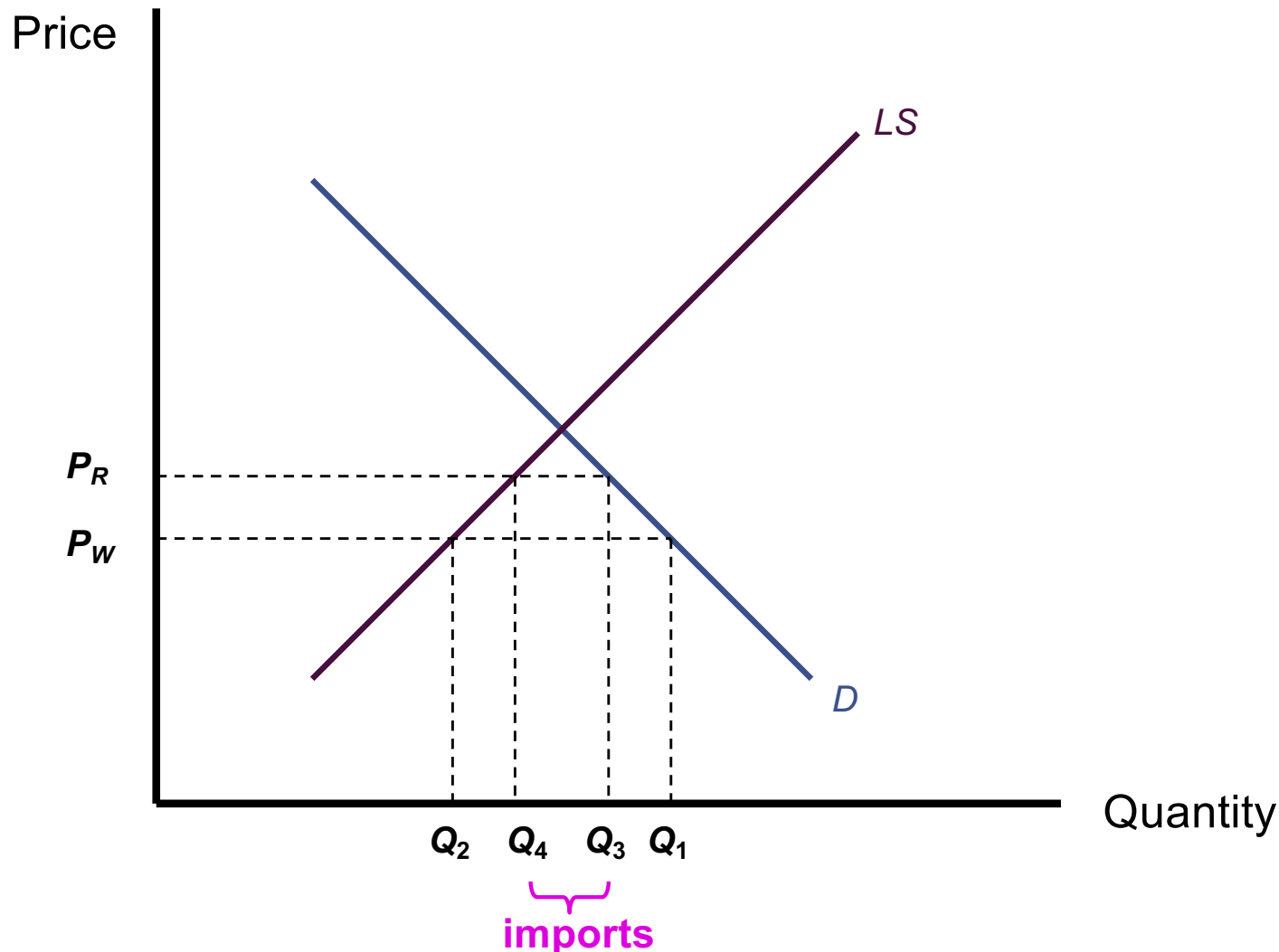
7. Economic Efficiency and Welfare Analysis

Example: Gains from international trade



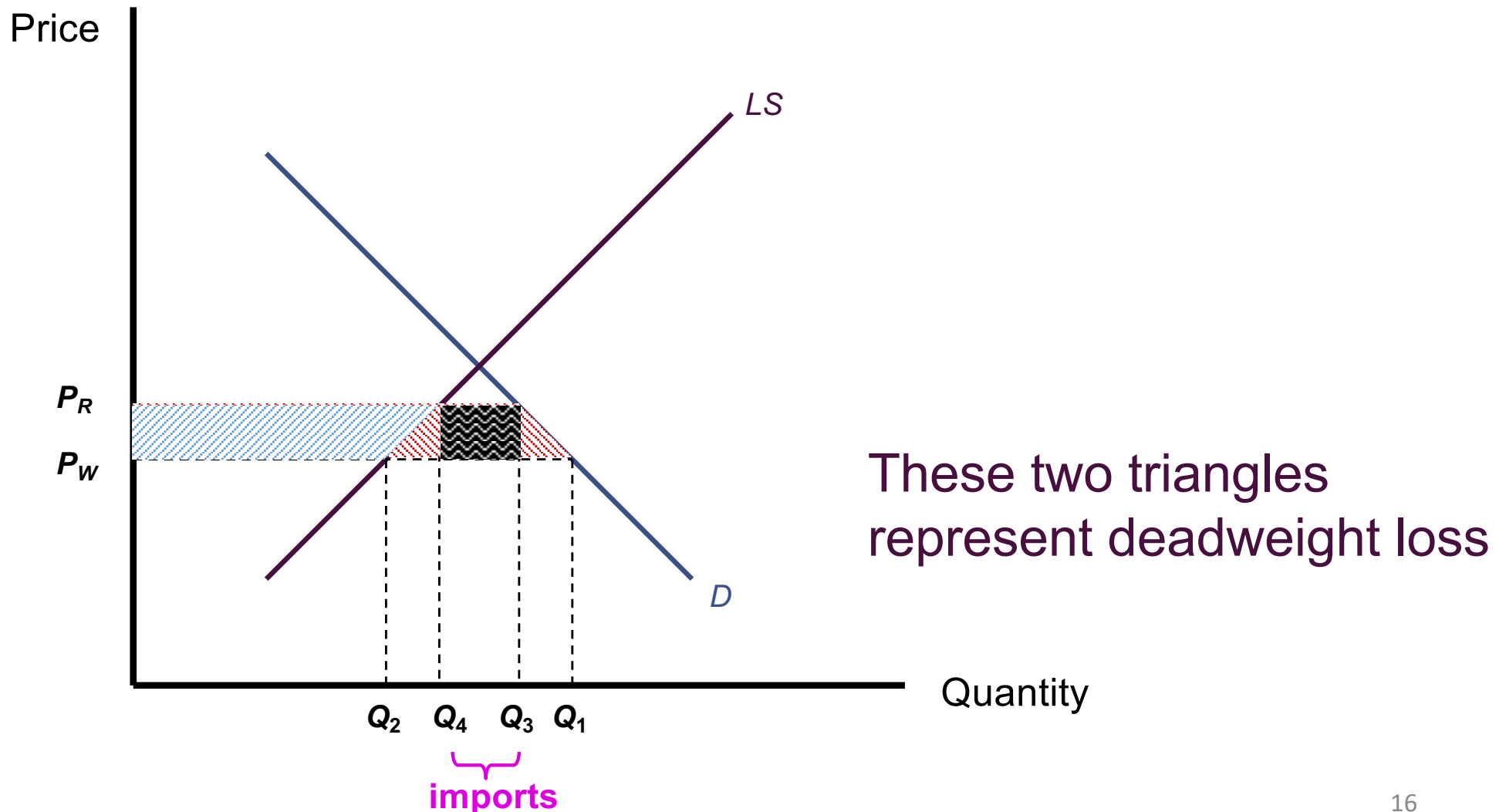
7. Economic Efficiency and Welfare Analysis

Example: Tariff protection and the politics of trade: $P_R = P_W + t$



7. Economic Efficiency and Welfare Analysis

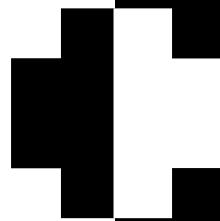
Example: Tariff protection and the politics of trade: $P_R = P_W + t$



7. Economic Efficiency and Welfare Analysis

Please review chapter 12 (The Partial Equilibrium Competitive Model) the mathematical models for these examples, and how to calculate the deadweight loss with the elasticity in the guide book Nicholson & Snyder

**Consumer
Theory**



**Producer
Theory**

Welfare Economics and Competitive Markets

Market Failures

Game Theory

General Equilibrium Model

Competitive Markets

General Equilibrium and Welfare

Partial equilibrium models of perfect competition are **inadequate** for describing all the effects that occur **when changes in one market have repercussions in other markets.**

Also, they are inadequate for making welfare statements about how well market economies perform → we need an **economic model that permits us to view many markets simultaneously.**

8. Perfectly Competitive Price System

We will assume that all markets are the way we described our one market in the partial equilibrium case → such a set of markets are a **perfectly competitive price system**.

We will assume that there is some **large number of homogeneous goods** in this simple economy (including the factors of production)

Each good has its **equilibrium price**, established by the action of supply and demand.

There are **no transaction or transportation charges**, and both (individuals and firms) have **perfect knowledge** of the market prices.

8. Perfectly Competitive Price System

The law of one price: since we assumed zero transactions cost and perfect information → **a homogenous good trades at the same price** (it doesn't matter who buys it or sells it)

Behavioral assumptions

- A large number of people buy any one good, where each one takes prices as given and adjusts her/his behavior to **maximize their utility** (given the prices and budget constraint). Someone may also be the “supplier” of productive services (labour), and they also regard prices (wages) as given
- With a large number of firms producing each good, each firm produces only a small share of the output. Firms are supposed to **maximize profits** and treat prices as given (for outputs and inputs)

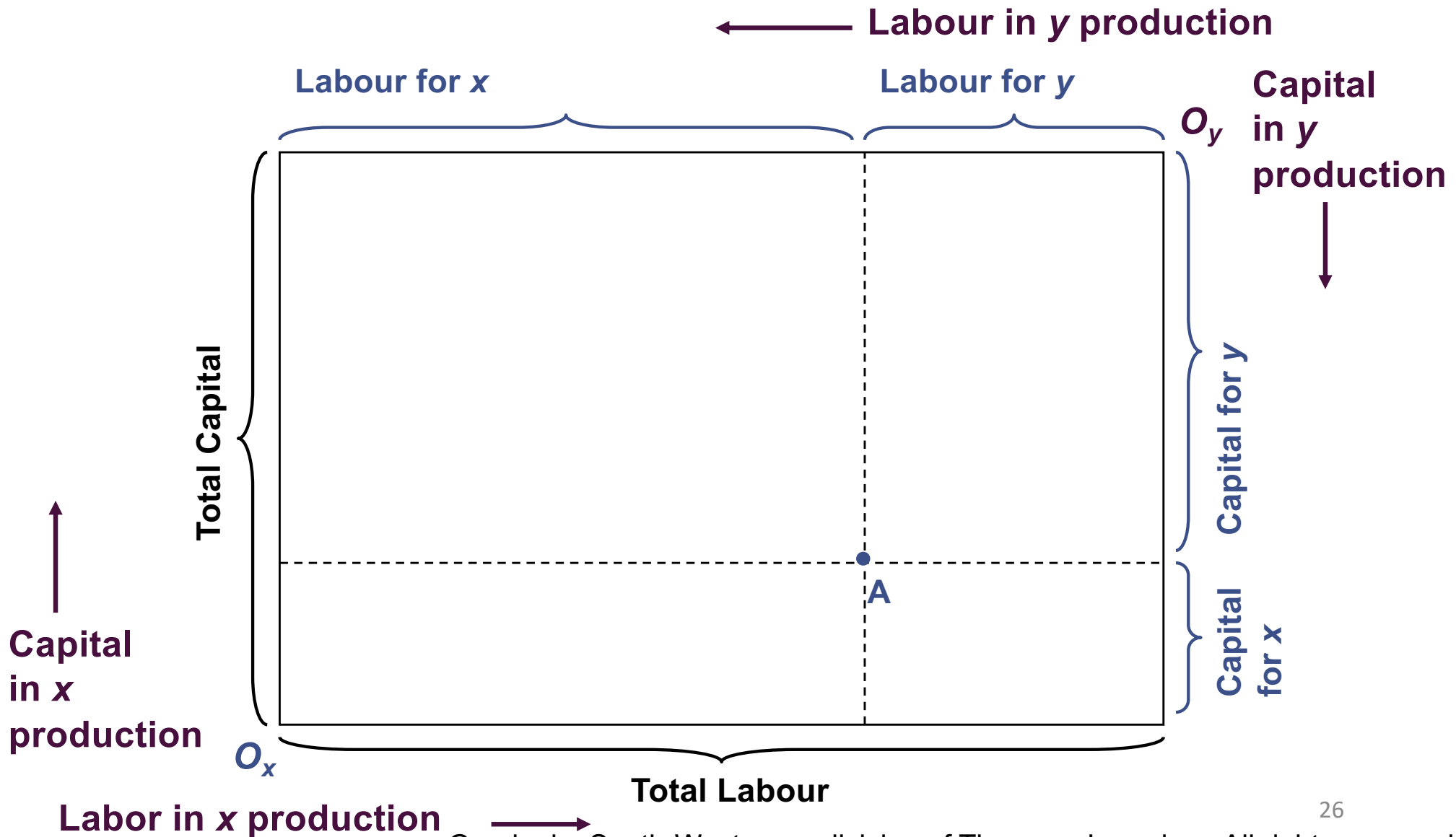
Model of General Equilibrium with Two Goods

Competitive Markets

9. General Equilibrium with Two Goods: Edgeworth Box

- Two goods $\rightarrow x$ and y
- General equilibrium demand:
 - Assume that all individuals have identical preferences, represented by an indifference curve map defined over quantities of x and y $\rightarrow$ show how individuals rank consumption bundles containing both goods
- General equilibrium supply
 - Use the production possibility curve $\rightarrow$ Edgeworth box diagram
 - We must assume that there are fixed amounts of capital and labour inputs that must be allocated to the production of these two goods
 - $\rightarrow$ The possible allocation of these inputs is shown in an Edgeworth box diagram, with dimensions given by the total amounts of capital and labour available

Edgeworth Box Diagram

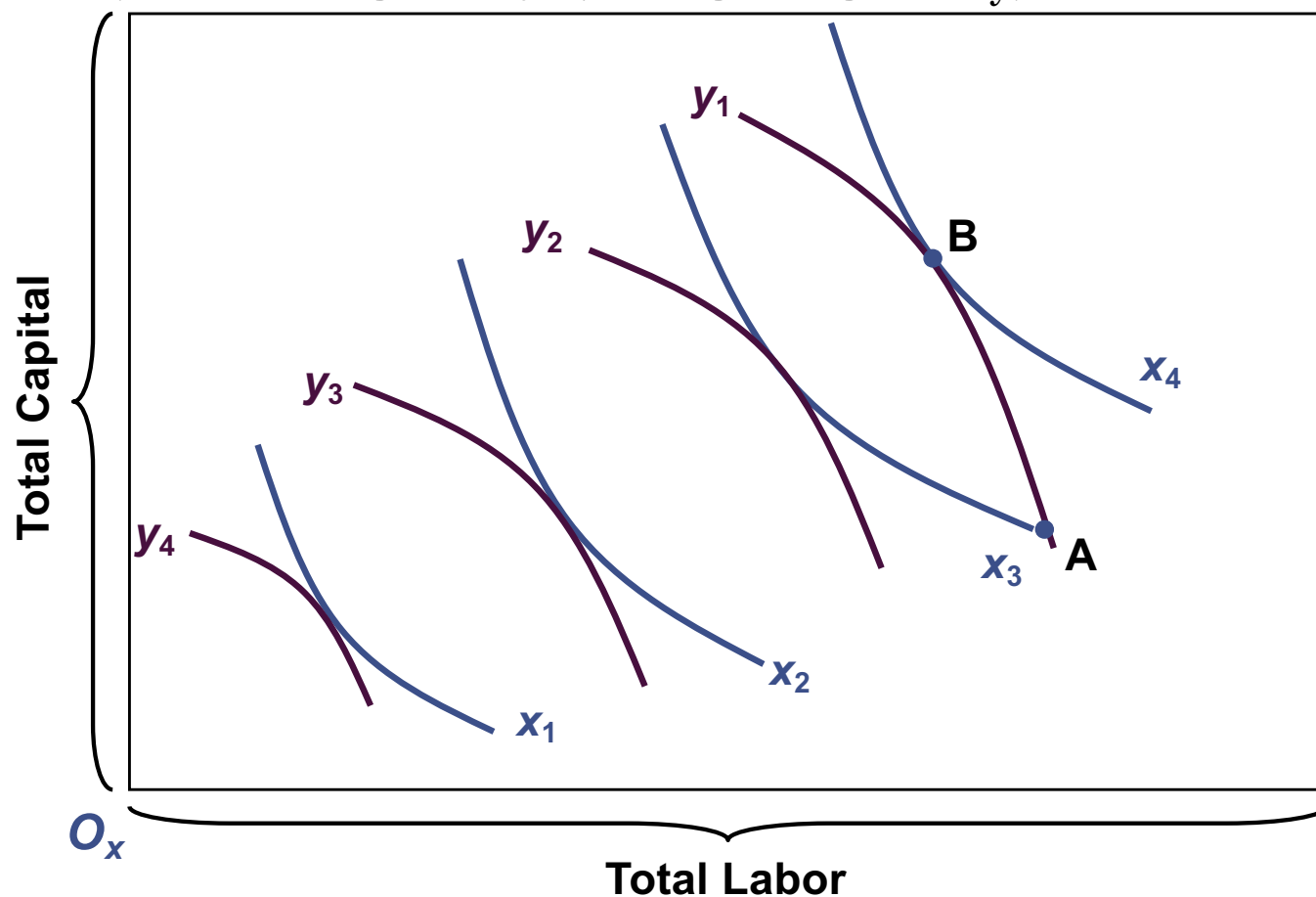


Edgeworth Box Diagram

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Efficient allocations

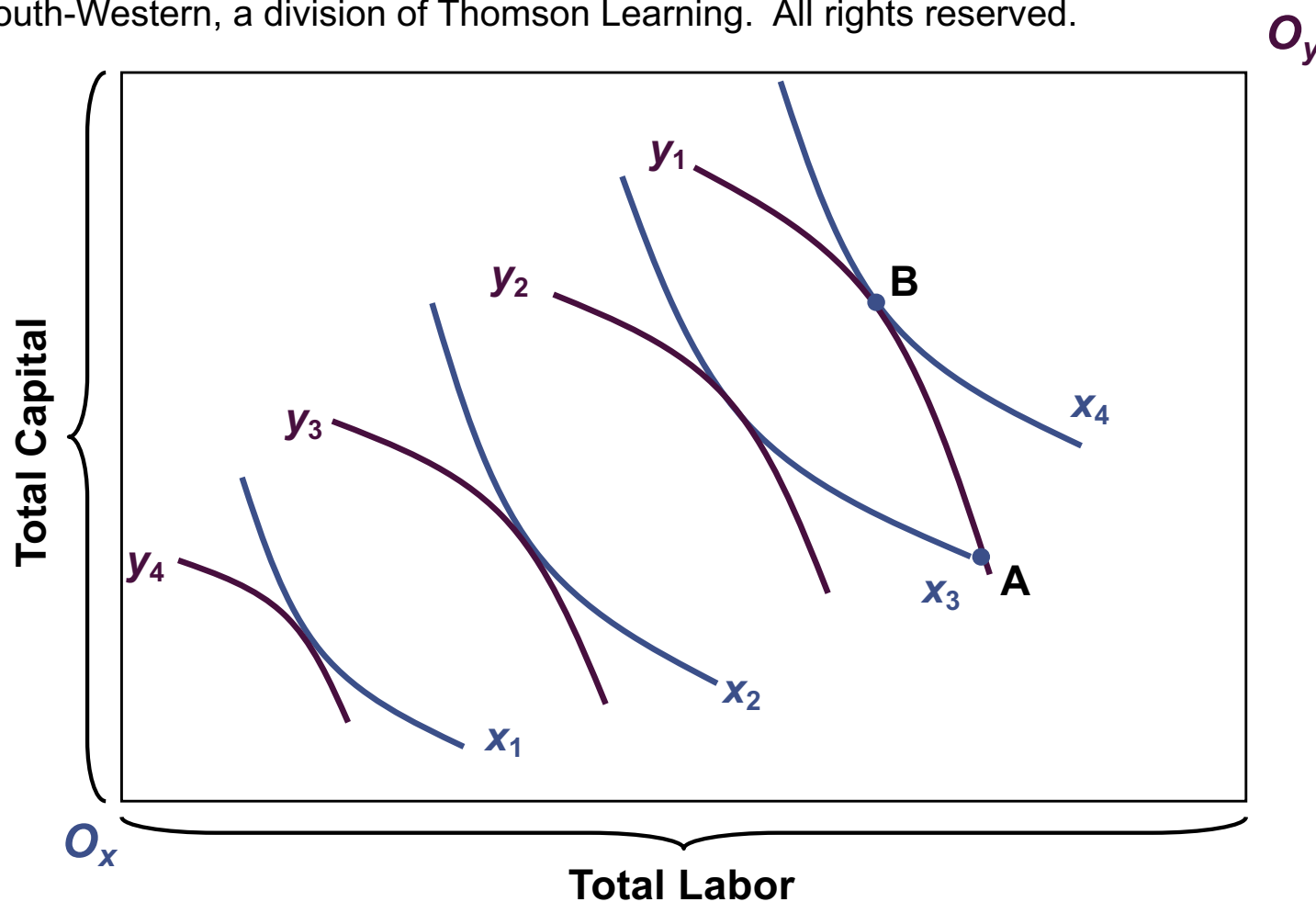
Many of the allocations shown in the Edgeworth Box are technically inefficient $\rightarrow$ we use isoquant maps for good x (using origin O_x) and for good y (using origin O_y)



Activity 2

Is point A efficient? Justify your answer.

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9. General Equilibrium with Two Goods: Edgeworth Box

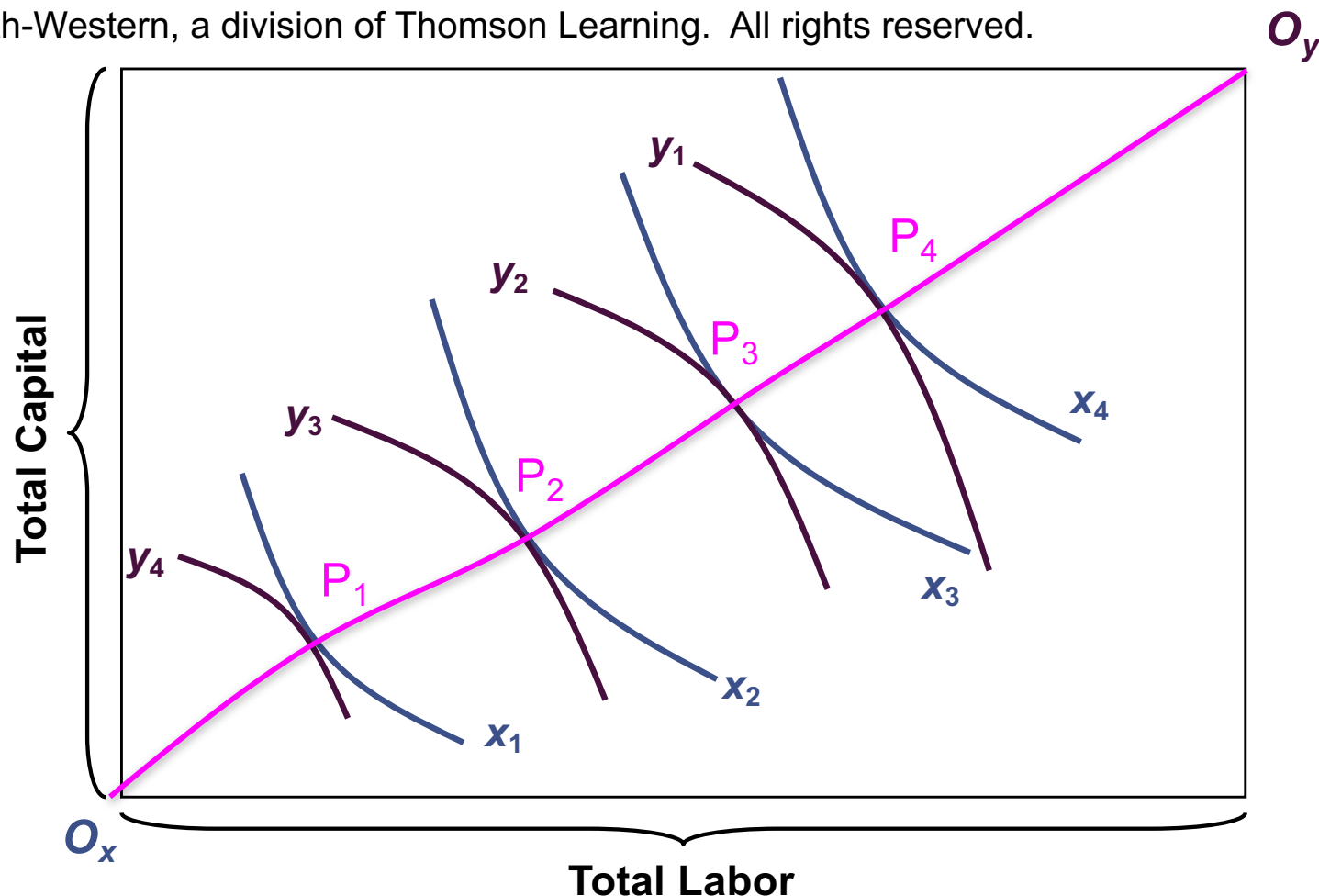
The efficient allocations shown are those such as P_1 , P_2 , P_3 , and P_4 , where the **isoquants are tangent to one another**.

At any other points in the box diagram, the two goods' isoquants will intersect, and we can show inefficiency as we did for point A. At the points of tangency, however, this kind of unambiguous improvement cannot be made. In going from P_2 to P_3 , for example, more x is being produced, but at the cost of less y being produced, so P_3 is not “more efficient” than P_2 —both of the points are efficient.

9. Edgeworth Box Diagram

Efficient locations are those such as P_1 , P_2 , P_3 and P_4 (isoquants are tangent to one another)

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9. General Equilibrium with Two Goods: Edgeworth Box

The line joining O_x and O_y (purple line) is the **locus of these efficient points**. Some texts call it the **contract curve**.

Along this line, when the isoquants are tangent, the RTS (of l for k) in the production of good x is equal to the RTS in the production of y .

$$\frac{w}{r} = RTS \text{ (of } l \text{ for } k)_x = RTS \text{ (of } l \text{ for } k)_y$$

9. General Equilibrium with Two Goods: Edgeworth Box

The Edgeworth Box is also used to model trade between two people since people who voluntarily trade benefit from that trade.

In this case, each person has an **endowment**: the initial allocation of goods.

The Edgeworth box represents this trade through their indifference curves, where the axes represent the combined endowments for each good.

Example: Edgeworth box in Trading with two people

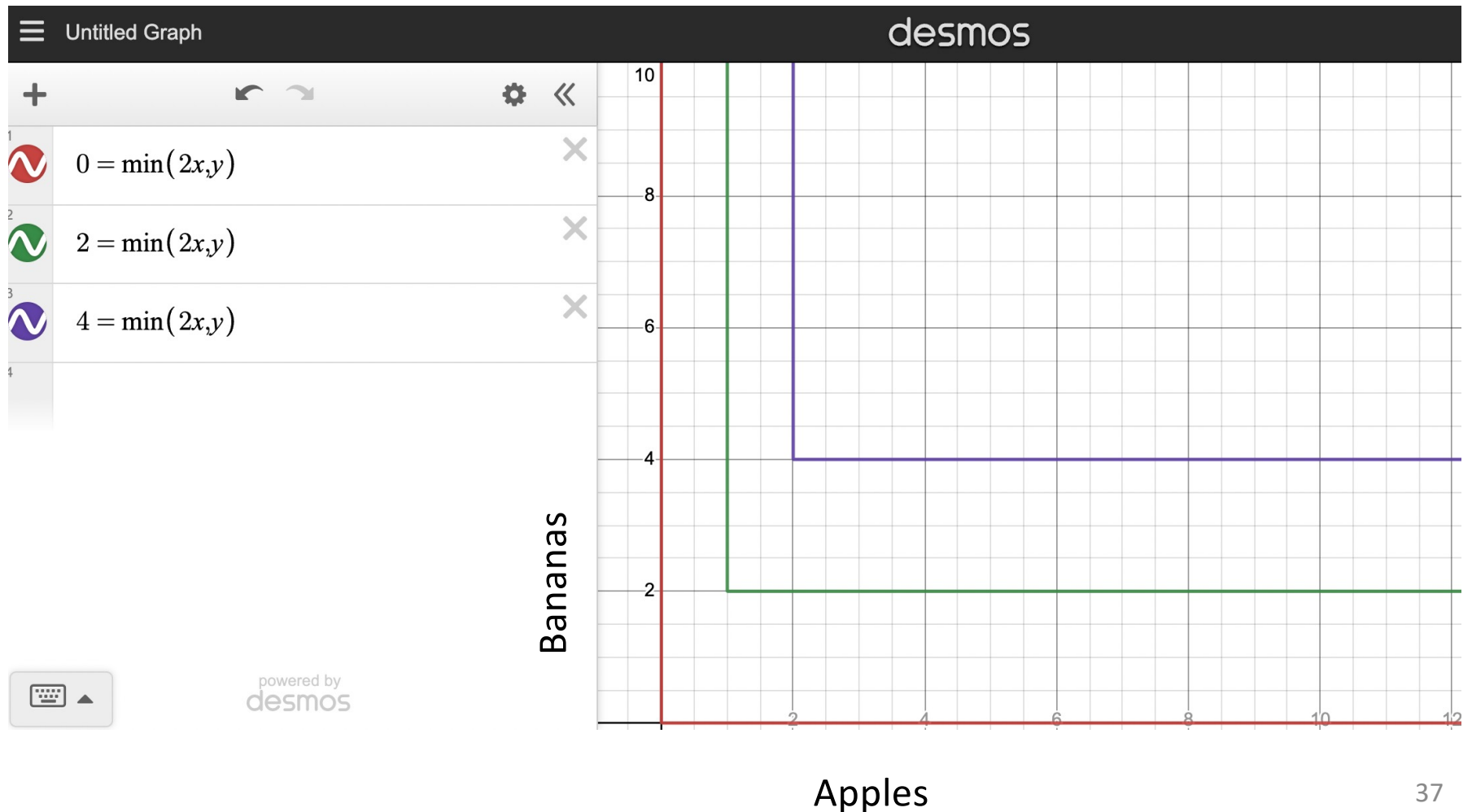
Lucy's utility function is $U(a, b) = \min\{2a, b\}$, where a is the number of apples she consumes, and b is the number of bananas she consumes. Linus' utility function is $U(a, b) = 2a + b$. Lucy has an initial endowment of 12 apples and no bananas, whereas Linus has an initial endowment of no apples and 12 bananas.

- a) Draw the Edgeworth box diagram, showing the initial allocation (use point E to denote it) and sketch a few indifference curves for Linus and Lucy.
- b) Show the locus of the Pareto optimal allocations and find the ratio of prices in a competitive equilibrium.
- c) How many apples and bananas will Lucy and Linus consume in competitive equilibrium? Think about their budget constraints.

Hint: Before writing your answer, test some allocations first

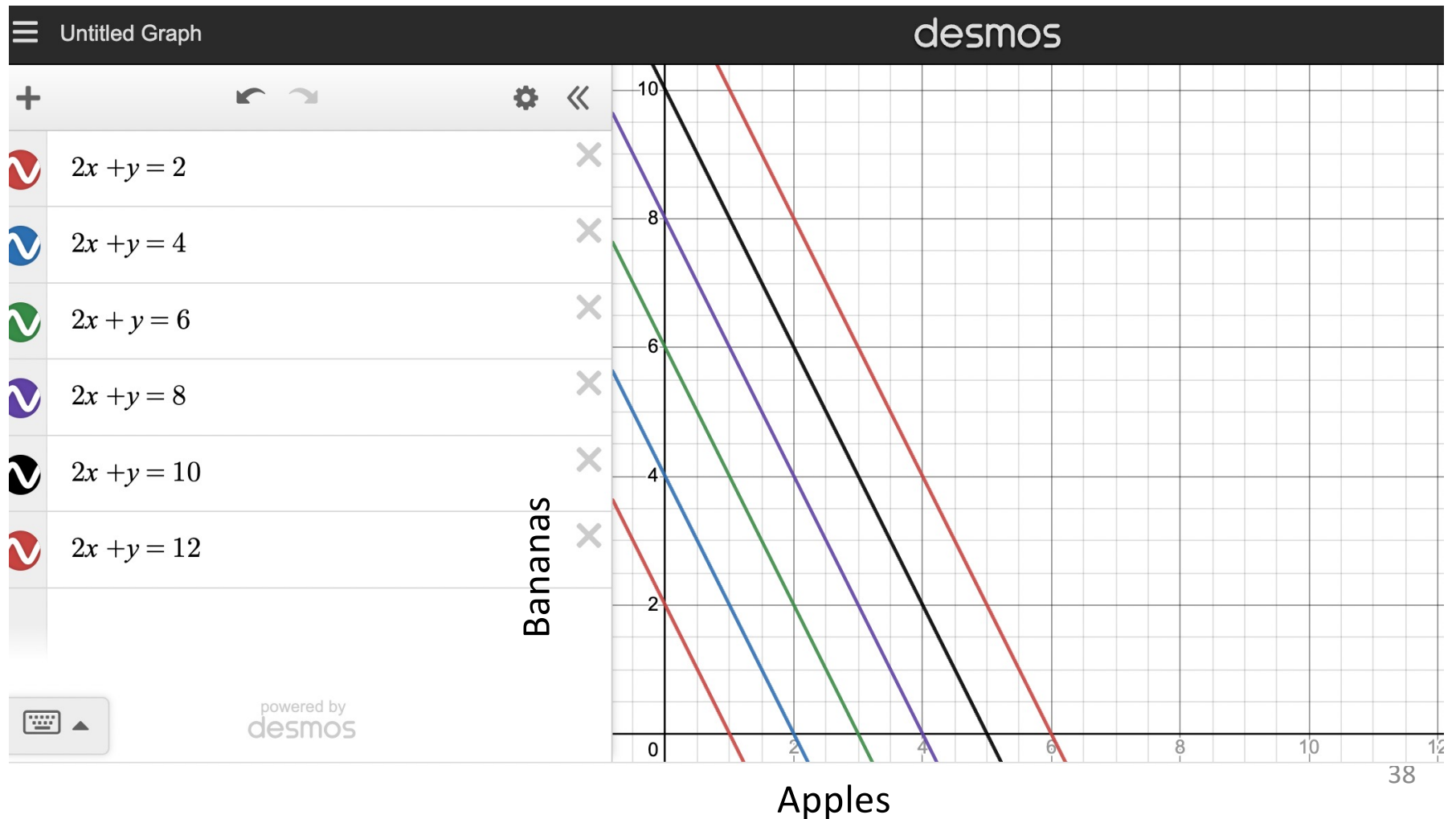
Example: Edgeworth box

Lucy's indifference curves.

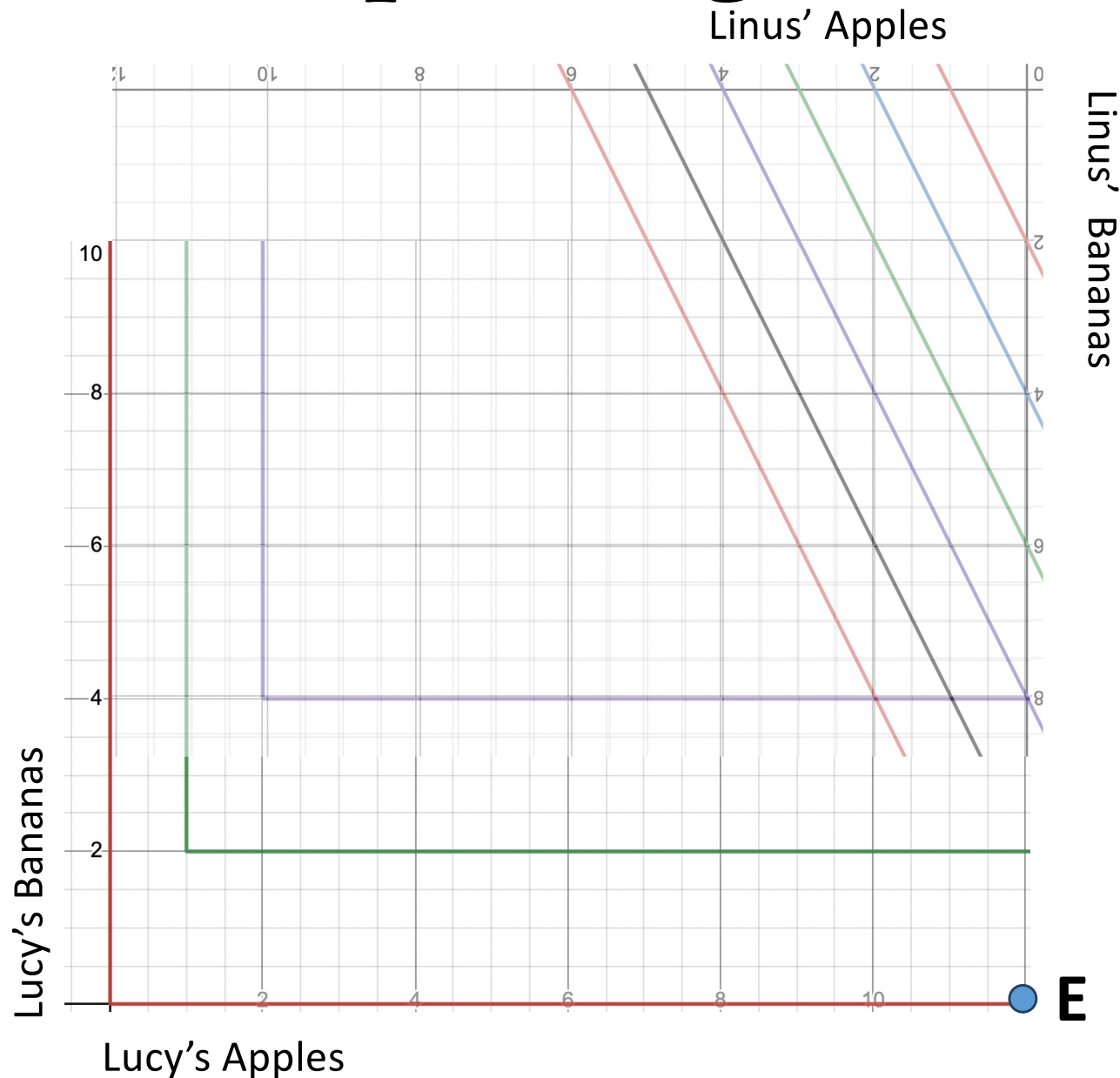


Example: Edgeworth box

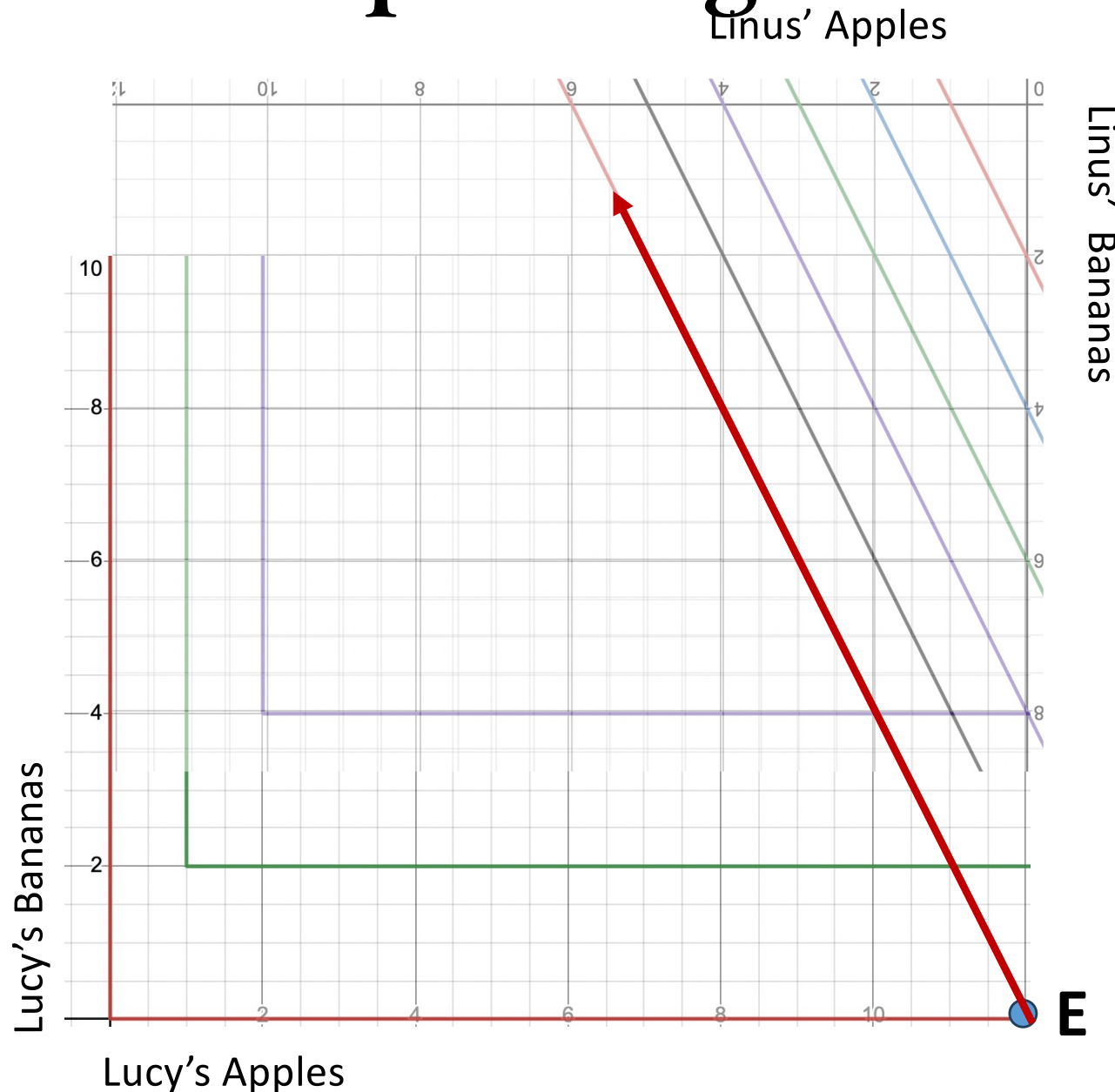
Linus' indifference curves.



Example: Edgeworth box



Example: Edgeworth box



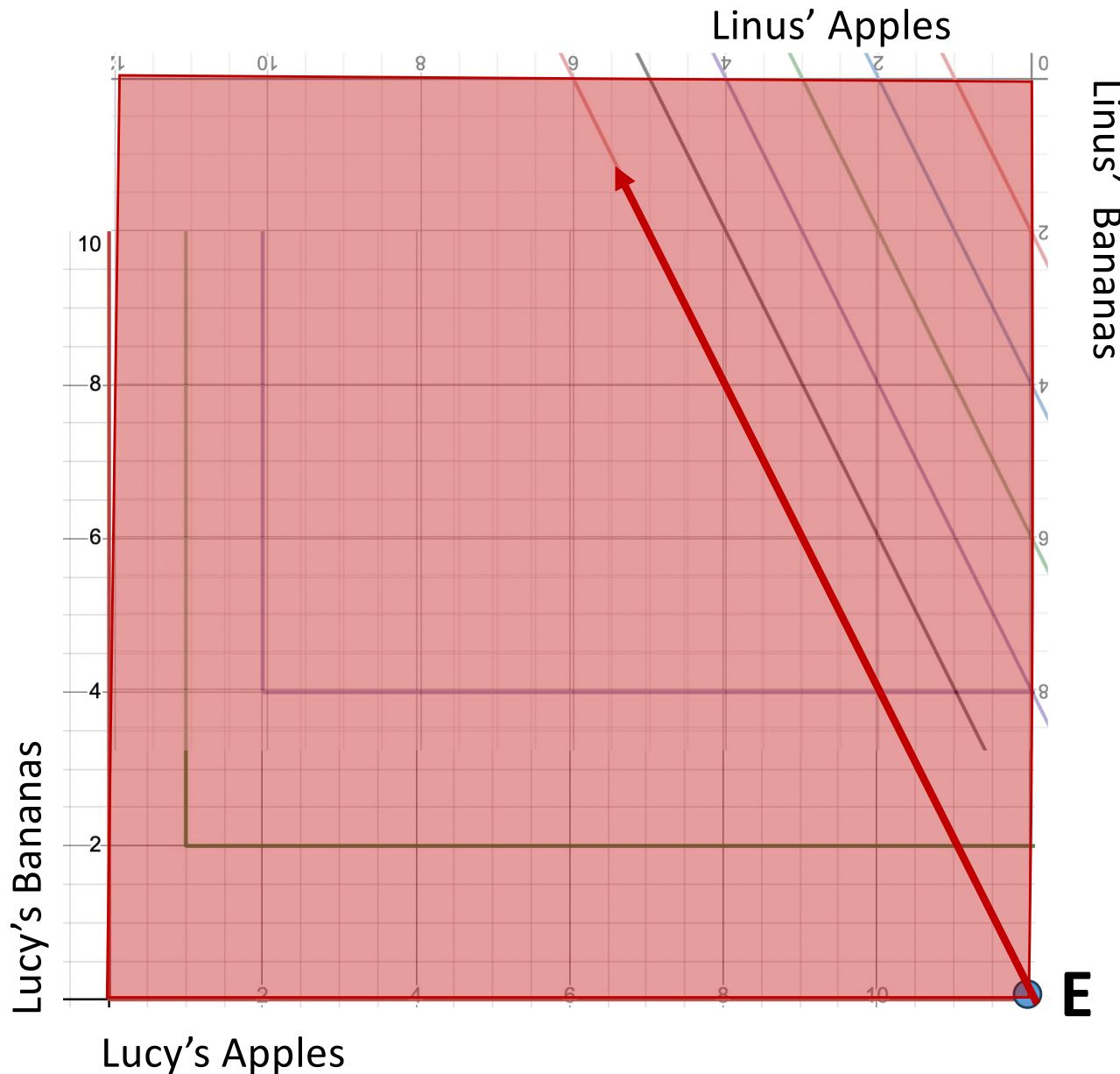
At the initial allocation (E):

$$U_{Lucy} = \min(2 \cdot 12, 0) = 0$$

$$U_{Linus} = 2 \cdot 0 + 12 = 12$$

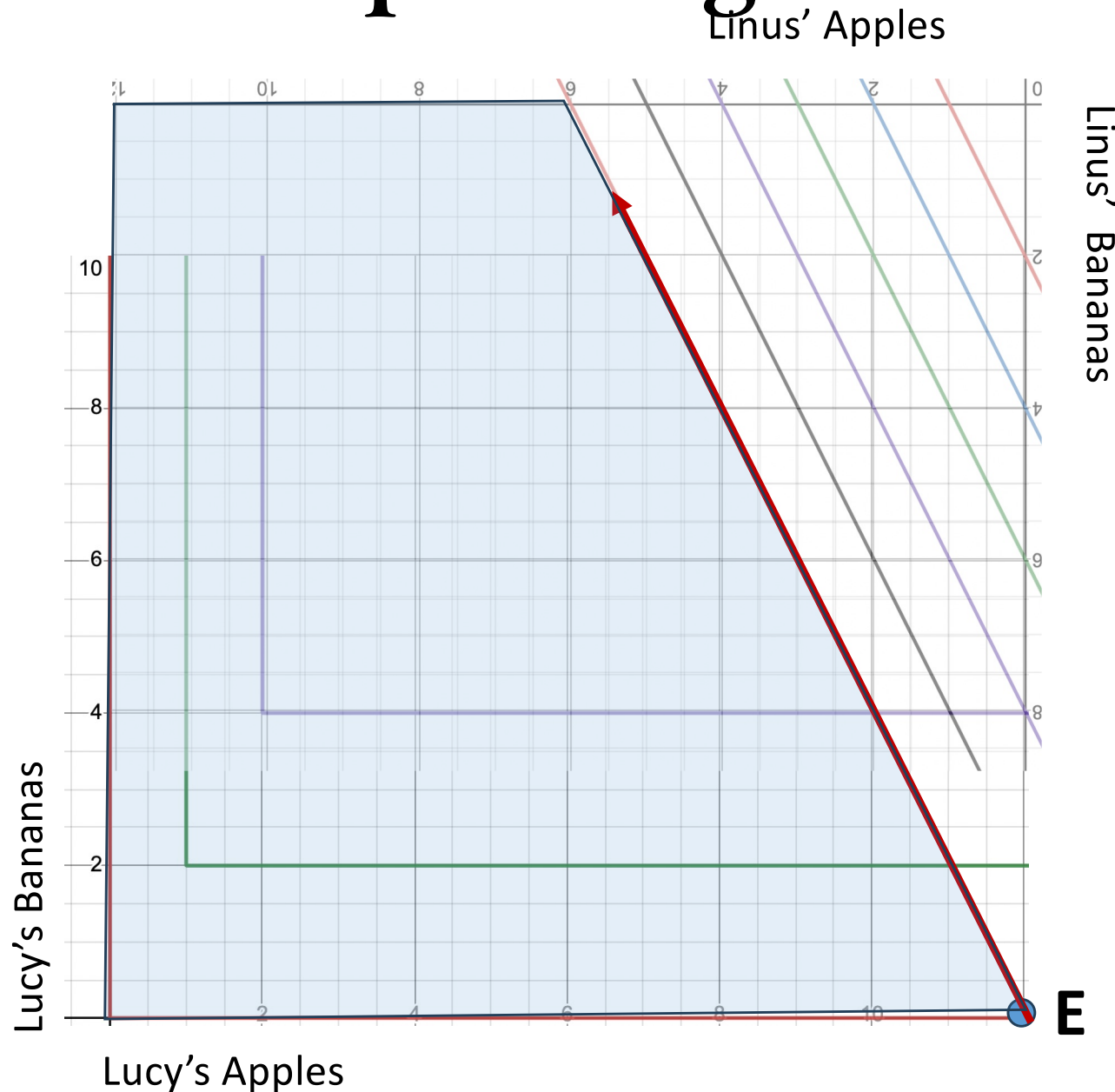
→ Lucy needs to trade some apples to get some bananas and improve her level of utility.

Example: Edgeworth box



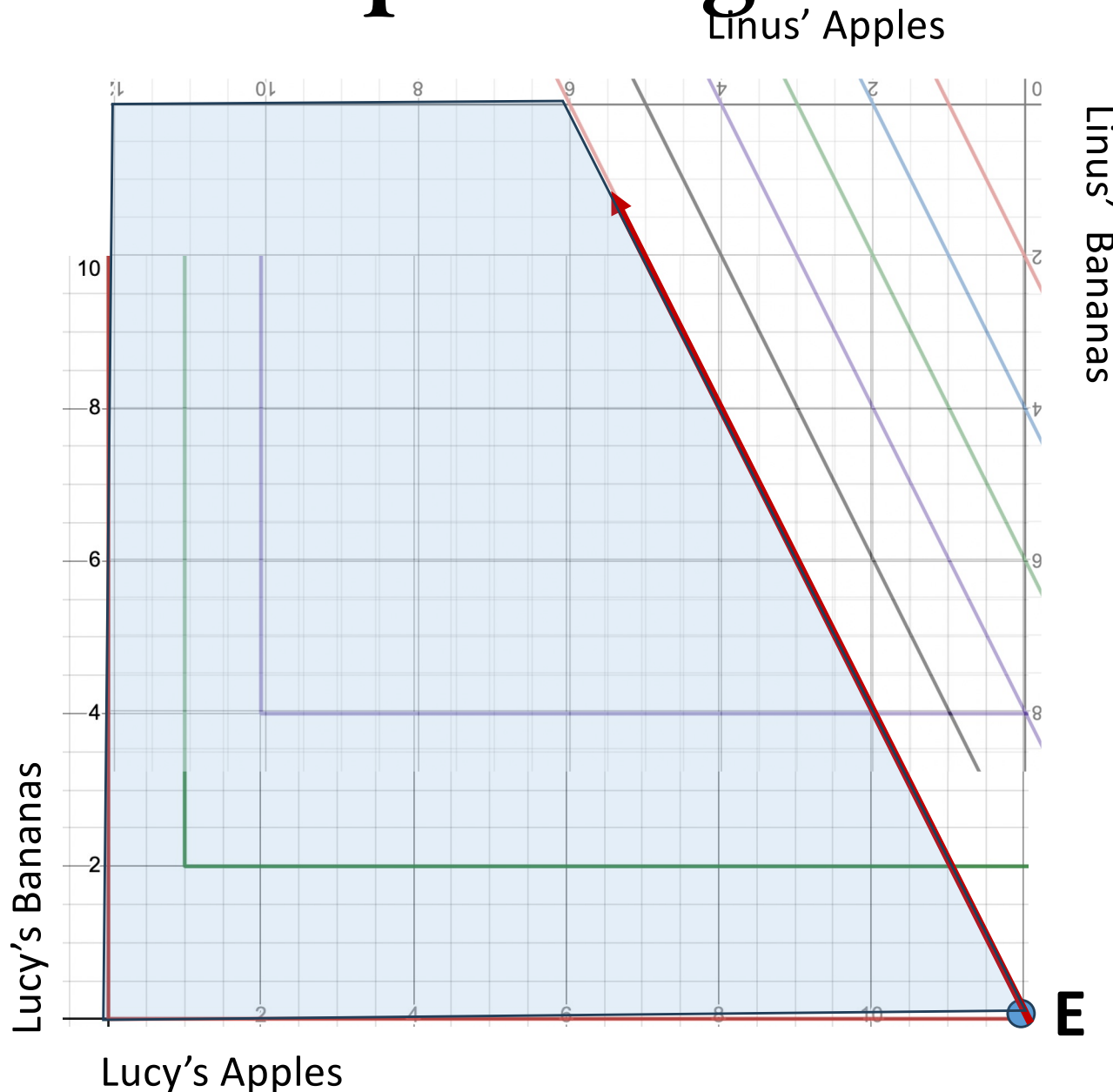
For Lucy, all the shadowed allocations are strictly preferred over the current allocation E.

Example: Edgeworth box



For Linus, all the shadowed allocations are strictly preferred over the current allocation E

Example: Edgeworth box

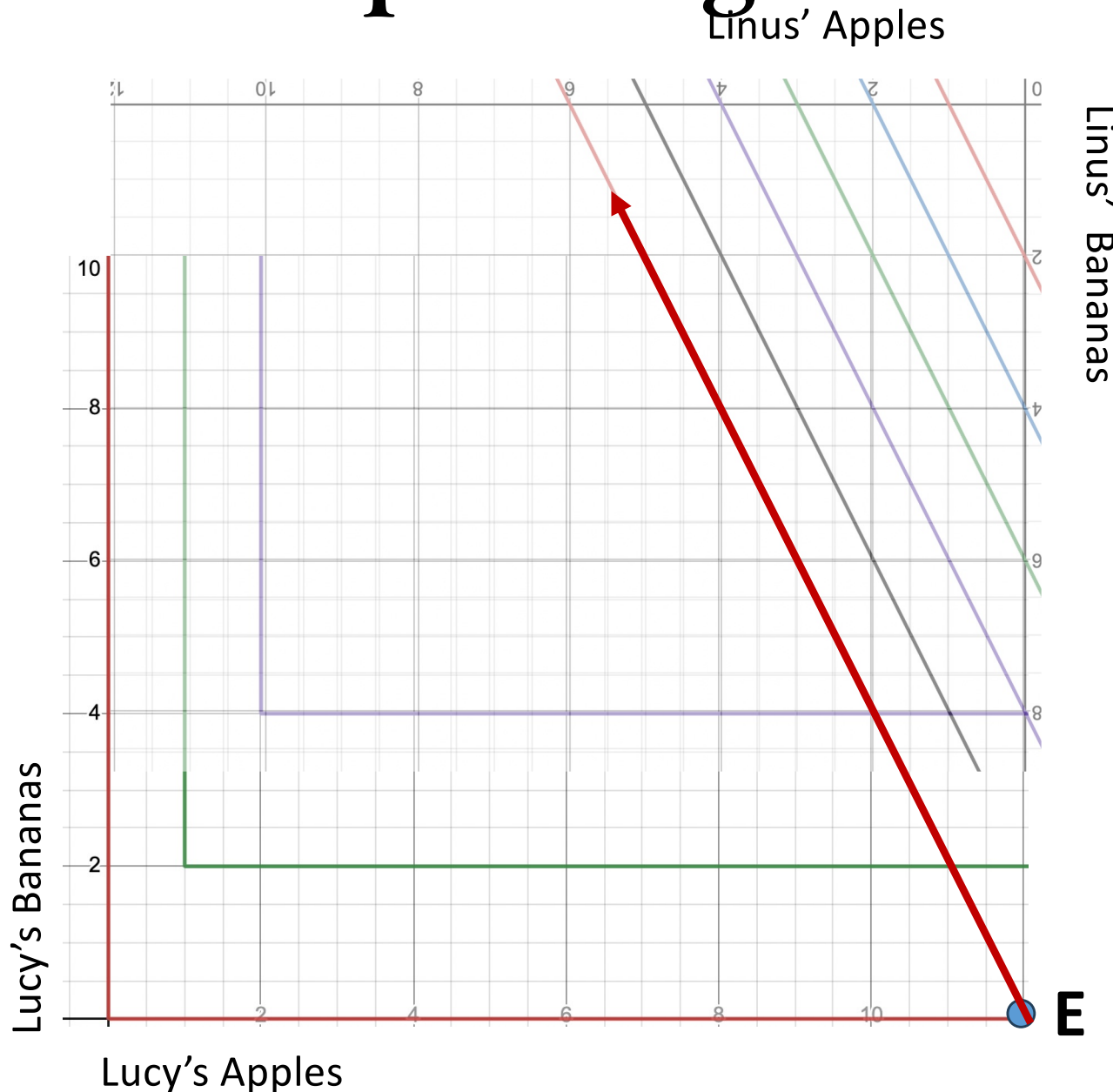


They need to **Exchange**, improving the level of satisfaction of one of them and maintaining the level of satisfaction of the other one.

Linus will be willing to trade as long as he is not worse off. He is indifferent if he receives one apple and gives two bananas in exchange → he will be moving in his indifference curve.

$$\frac{P_{apples}}{P_{bananas}} = \frac{2}{1}$$

Example: Edgeworth box



At the initial allocation (E):

$$U_{Lucy} = \min(24, 0) = 0$$

$$U_{Linus} = 2 \cdot 0 + 12 = 12$$

When Lucy gives 1 apple for 2 bananas

$$U_{Lucy} = \min(22, 2) = 2$$

$$U_{Linus} = 2 \cdot 1 + 10 = 12$$

When Lucy gives another apple for 2 bananas

$$U_{Lucy} = \min(20, 4) = 4$$

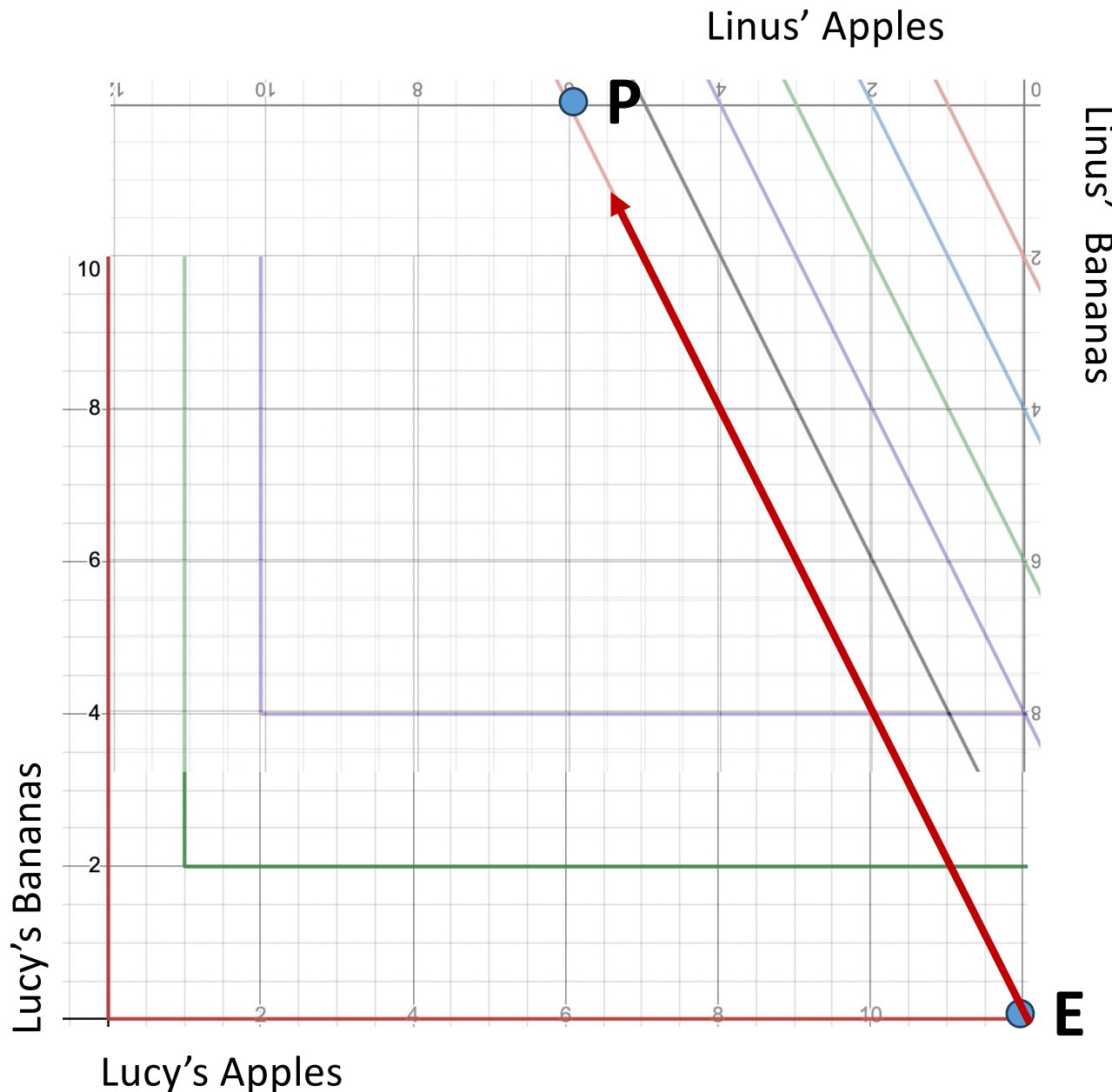
$$U_{Linus} = 2 \cdot 2 + 8 = 12$$

When Lucy gives another apple for 2 bananas

$$U_{Lucy} = \min(18, 6) = 6$$

$$U_{Linus} = 2 \cdot 3 + 6 = 12$$

Example: Edgeworth box



When Lucy gives another apple for 2 bananas

$$U_{Lucy} = \min(16, 8) = 8$$

$$U_{Linus} = 2 \cdot 4 + 4 = 12$$

When Lucy gives another apple for 2 bananas

$$U_{Lucy} = \min(14, 10) = 10$$

$$U_{Linus} = 2 \cdot 5 + 2 = 12$$

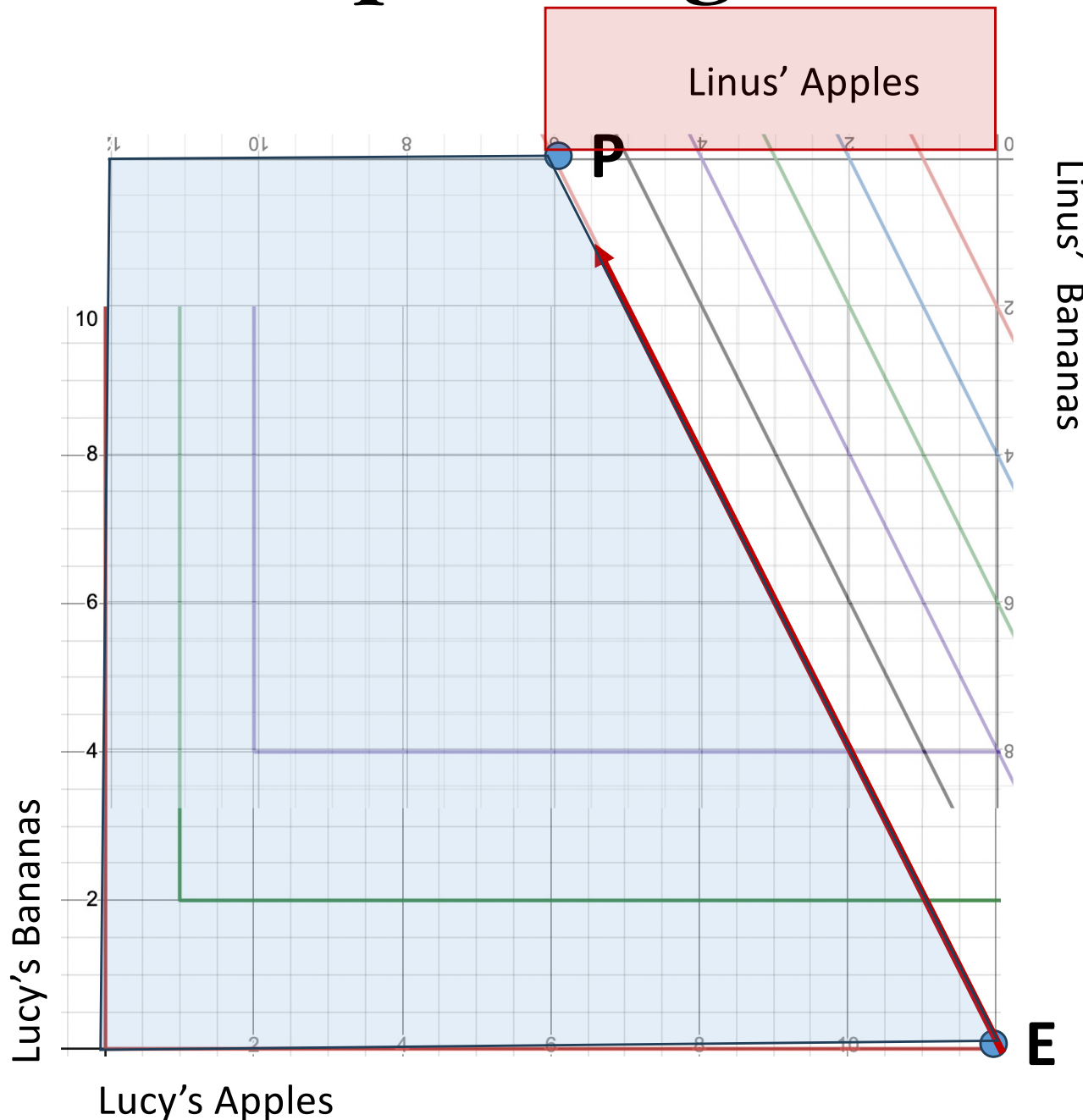
When Lucy gives another apple for 2 bananas (point P)

$$U_{Lucy} = \min(12, 12) = 12$$

$$U_{Linus} = 2 \cdot 6 + 0 = 12$$

Here, they stop because Linus has no more bananas and if Lucy exchanges more apples for bananas, she will obtain a lower utility level.

Example: Edgeworth box



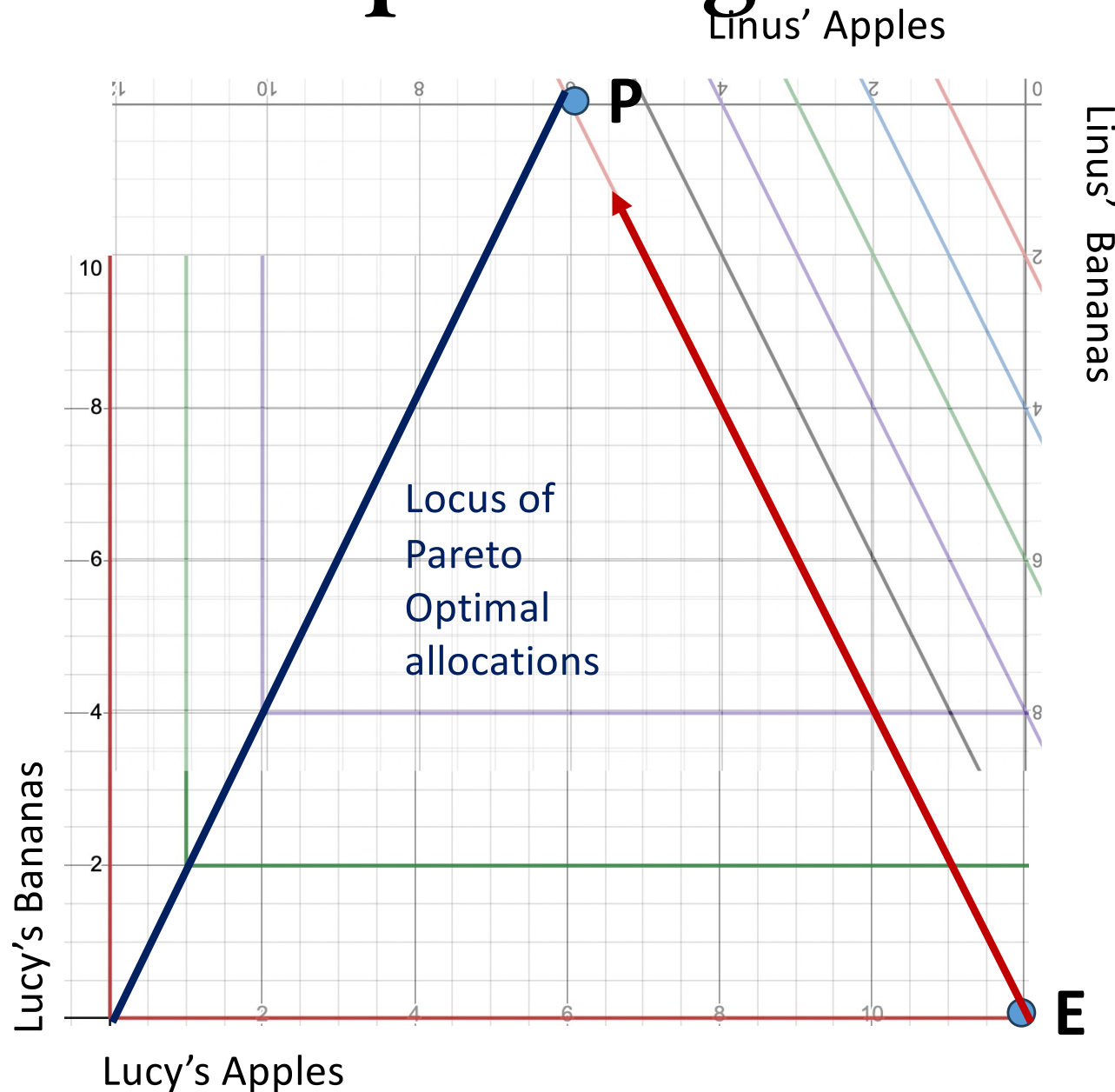
What happens at P?

The points that make Linus better off (taking point P as a reference) are in the light blue area.

The points that make Lucy better off are in the red area.

These two areas do not intersect each other → P is an efficient point.

Example: Edgeworth box



The **blue line** depicts the locus of pareto optimal allocations.

9. General Equilibrium with Two Goods: Edgeworth Box

As in the previous Edgeworth box, the **locus of these Pareto optimal allocations** can be called the **contract curve**.

Along this line, when the indifference curves are tangent, the MRS between these two people should be equal:

$$\frac{P_x}{P_y} = MRS_{Person\ 1} = MRS_{Person\ 2}$$

Individual Activity

In a piece of paper write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it, in your own words.

If you have any **questions** about we discussed today, write it down too. We will go through them at the beginning of next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.